

AKSHAT JAISWAL

Senior Software Engineer | Backend & Distributed Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 3.5+ years of experience architecting high-throughput distributed FinTech systems and backend services using Java and Spring Boot. Strong expertise in Java Virtual Threads (Project Loom), Redis distributed locking, and zero-downtime data migrations. Proven track record of scaling identity pipelines to 120,000+ daily transactions (1.5M+ API calls), executing live production risk management, and mentoring engineering pods.

TECHNICAL SKILLS

- **Languages & Frameworks:** Java (8 / 17 / 21 / 25), Java Virtual Threads (Project Loom), Spring Boot, Spring Security (OAuth2 / RBAC), Spring Data JPA, Hibernate, Apache PDFBox 3.x, SQL
- **Distributed Systems & Cloud:** Redis (Distributed Locking SET NX, Caching), AWS (S3, RDS, EC2, CloudWatch, SNS), Microservices Architecture, Event-Driven Systems
- **Databases & Data Operations:** MySQL 8.0 (Index Tuning, Query Optimization, Multi-DB Routing), Batch Migrations, High-Throughput Data Purge Engines
- **FinTech, Security & Testing:** Aadhaar eSign (CCA / IT Act 2000), NPCI E-NACH, DigiLocker APIs, RBI Video-KYC (VKYC), UIDAI OVSE, JWE Encryption, VAPT Remediation, Docker, Git, CI/CD, JUnit 5, Mockito

PROFESSIONAL EXPERIENCE

Digitap.AI | Bengaluru, India

Aug 2023 – Present
Senior Software Engineer

Engineering Leadership, Incident Command & Risk Management

- Lead and mentor a pod of 5 engineers (2 Full-Time SEs, 3 Interns); drive system architecture reviews, sprint backlog planning, and defensive coding standards across core banking and identity rails.
- Serve as primary on-call incident commander for production-only rails (DigiLocker, OKYC, E-NACH, eSign); rapidly diagnose upstream bank/UIDAI spec drifts and roll out zero-downtime hotfixes to mitigate Sev-1 partner disruptions.

Digitap eSign Gateway v2 (Core Architecture & Contract Rail) | Java 25, Virtual Threads, Redis, PDFBox 3.x

- Architected enterprise digital contract execution gateway from scratch on Java 25 & Virtual Threads, delivering sequential (distributed lock-enforced) and parallel signing workflows with in-memory PDF processing.
- Slashed callback round-trip latency from ~550ms to <50ms by eliminating synchronous REST calls through direct in-memory Base64 decoding from TSP callbacks.
- Built in-memory multi-document coordinate stamping using Apache PDFBox 3.x for single-OTP multi-file signing; added upload-time pre-flight validation to prevent mid-flow runtime crashes.
- Designed transaction-aware webhook dispatcher with 6 lifecycle events, ensuring idempotent client notifications across retries using Redis-based deduplication (SET NX EX).

DigiLocker Revamp & Data Lifecycle Platform | Spring Boot, Virtual Threads, MySQL, AWS S3/SNS/CloudWatch

- Re-architected DigiLocker infrastructure by consolidating 4 legacy microservices into a unified modular application; expanded platform capacity to sustain 120,000+ daily successful transactions (~1.5M+ daily API invocations across a multi-step orchestration pipeline) while cutting p95 latency from 30s+ to 7–10s via concurrent Virtual Threads.
- Engineered boundary-based dual-DB pagination routing to transparently query legacy and new databases with unified pagination, avoiding historical data migration downtime.
- Developed an asynchronous multi-threaded data purge engine operating at 2,500 records/sec against AWS S3 and RDS, purging 12TB+ of obsolete KYC payloads and reducing cloud storage costs by ~40%.
- Configured AWS CloudWatch log metric filters and operational dashboards, reducing Mean Time to Detection (MTTD) for downstream gateway degradation by 60%.

VKYC V2 & Security Infrastructure | Spring Boot, Spring Security, Redis, JWE, VAPT

- Implemented core security framework with Spring Security RBAC across 4 operational roles (Agent, Admin, Lead, QA) and resolved 15+ critical VAPT vulnerability findings to comply with RBI data protection mandates.
- Integrated JWE (JSON Web Encryption) payload encryption and OAuth2 token management for enterprise FinTech partners, ensuring end-to-end data privacy for sensitive customer KYC records.
- Overhauled agent assignment concurrency algorithms, dropping allocation collisions to <0.01% across peak concurrent call volumes for tier-1 enterprise lending partners.

Banking Rails & Regulatory Compliance (NACH / OVSE) | Spring Boot, AWS SNS, Crons

- Automated end-to-end physical and e-mandate banking operations with Kotak & Yes Bank via scheduled crons and AWS SNS parsers, eliminating manual reconciliation and reducing clearance turnaround from 3 days to real-time.
- Engineered a standalone UIDAI-compliant Offline Verification Seeking Entity (OVSE) XML verification service in a 7-day regulatory sprint with 99% automated test coverage using JUnit 5 and Mockito.

Merck Group | Bengaluru, India

July 2022 – July 2023
Software Developer Intern

- Engineered 15+ RESTful backend APIs for pharmaceutical catalog and order reconciliation services using Java 17 and Spring Boot, serving 10,000+ daily internal queries.
- Automated batch inventory synchronization pipeline using Spring Data JPA and MySQL batching, cutting nightly reconciliation lag by 35% across legacy enterprise databases.

HONORS & RECOGNITION

- **Rising Star Award (Digitap, 2024):** Conferred for top engineering impact in untangling high-liability legacy architectures into scalable microservices.
- **Quarterly Excellence Award (Digitap, 2025):** Recognized among 150+ engineers for execution velocity across DigiLocker consolidation and 7-day OVSE compliance rollout.

EDUCATION

Bachelor of Technology (B.Tech) in Information Technology | First Division (I-DIV) | Graduated 2022

Dr. A.P.J. Abdul Kalam Technical University (AKTU) / Buddha Institute of Technology